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Project Title:	The Dynamics of Heterogeneous Solidification Microstructure Genesis in Undercooled Composite Systems		
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Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
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	The ground based program represents a balanced experimental and analysis effort directed toward the investigation of drop tube, glass encased and levitation containerless processing methods. The investigation focus is on the understanding and analysis of microstructural evolution during solidification of undercooled melts and composite systems. The liquid undercooling level in a laboratory scale (3 m) drop tube and levitation melting system can be altered through the variation of processing parameters such as alloy composition, melt superheat, sample size and gas environment. The solidification behavior is evaluated through metallography, SEM, TEM, DTA, and x-ray diffraction in conjunction with a heat flow model of the processing conditions to judge the sample thermal history. A central component of the current program addresses the use of containerless processing together with an undercooled melt with incorporated particles to develop a critical evaluation of solidification front-particle interactions. Another component of the current investigation		

Task Description:

is the analysis of the control of microstructural evolution in Ni-V, Nb-Pd and Co-Al alloys through the high undercooling levels provided by containerless processing to provide fundamental new information on the control of nucleation. The microgravity environment provides new opportunities for the identification and characterization of key parameters that control of solidification processes, and the containerless environment afforded by microgravity conditions avoids crucible-induced sample contamination and allows for large undercooling. The main research objective is the evaluation and analysis of the undercooling and resultant solidification microstructures in containerless processing, including drop tube processing and levitation melt processing of selected alloys and composites. Containerless processing in ground-based drop tubes simulates microgravity conditions via solidification of liquid samples under free fall conditions. The containerless environment is also attained in levitation melt processing and removes a major source of impurities and heterogeneous nucleation sites, allowing for a large melt undercooling. This enhanced liquid undercooling exposes alternate solidification pathways, allowing for the formation of novel microstructures. Controlling the undercooling level provides some control of the operative solidification pathway and the resultant microstructure. The novel structures that may be produced in a ground-based containerless processing facility preview the wide range of possible materials processing experiments that may be conducted in a space-based laboratory. The results of the ground-based study will be used to identify critical experimental variables in microgravity processing and the analysis can be used to design and define the science and hardware requirements for extended duration space experiments. In the solidification of composites the interaction between the solidification front and particles is central to effective processing. Undercooled liquid solidification simulates convection-free solidification conditions for such composites in a 1g environment. Numerous models for the interaction behavior have been proposed for planar solidification fronts, but critical tests are lacking in many cases due to confounding gravitational effects. The interaction of a fast-moving, dendritic solidification front with dispersed particles during solidification processing of metal-matrix composites has revealed a new mode of the incorporation behavior that is not included in existing models for planar interfaces and has not been experimentally accessed previously. In order to complete the analysis of the new microstructural class developed during duplex partitionless solidification it is important to address the key issue of independent or coordinated nucleation of the product phases. Experiments on undercooled samples have provided the microstructural characteristics that are essential to answer this issue. With the critical information on the nucleation mechanism a complete kinetics model analysis can be formulated for the duplex partitionless reaction. The outcome of this program will provide a unique background for a space experiment, because never before has it been possible to quantify the solidification kinetics and the associated microstructure evolution in a manner that can be achieved with the proposed experiments.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

The current studies have established that some pure metals and alloys can exhibit large undercooling levels in the presence of crystalline particles. A wide range of growth velocities can be examined through variation of the undercooling level. Highly undercooled samples provide an excellent means to study the interaction of high-velocity dendrites and particles and test theoretical models over an extended range of dendrite tip velocities. Even in samples at modest undercooling levels (on the order of 50 K), the dendrite tip velocities of pure copper and nickel are about 5 orders of magnitude larger than the settling velocities for the matrix-particle systems in the current studies. Consequently, the solidification front interacts with a snapshot of the spatial particle distribution in the liquid, minimizing dynamic effects due to particle motion (due to settling or convective fluid flow) during solidification. Another set of experiments is focused on the examination of the transitions between particle incorporation modes (i.e. pushing, entrapment, and engulfment). Previous experiments have shown that there is a critical interface velocity (corresponding to a critical undercooling level) required to engulf particles in nickel-tantalum oxide and copper-tantalum oxide composites. The current studies have shown that the transition from entrapment to engulfment is not observed in the copper-aluminum oxide, copper-zirconium oxide, and nickel-zirconium oxide composite system at any experimentally accessible undercooling (up to 120 K). Samples with copper-nickel alloy matrix materials and zirconium oxide particles at similar undercooling levels (up to 160 K) also do not exhibit engulfment. In addition, the particle distributions within the alloy samples are similar to the particle distributions in pure copper and nickel matrix materials. These alloy studies indicate that density changes and solute diffusion do not significantly alter the particle incorporation behavior in highly undercooled samples.

Experiments are currently in progress to determine the effect of particle size and particle morphology on the distribution of particles in the microstructure of nickel-tantalum oxide and copper-tantalum oxide composite systems. The current studies have established that large undercooling levels can be achieved in the model composite systems and these undercooling levels are independent of the effect of particle size. The samples studied in this work undercool reproducibly by more than 100 K. The average particle diameter more than 10 micron- particles are confined around the sample in all undercooling levels. Experimental analyses have shown that the particles are distributed homogeneously inside the liquid sample. Thus, the confined particle around the sample is a result of particle movement produced by the moving solidification front. Furthermore, the particle size also alter the mechanism of grain refinement in an undercooled liquid. Detailed studies of the effects of interface velocity and alloy composition including composition that yield neutral buoyancy for particles and the effect of other experimental factors will be conducted to yield a clear understanding of the particle/solidification-front interaction that is needed to evaluate more complete analytical models.

To extend the current experiments and analyses a differential thermal analyzer has been constructed that is capable of nucleating an undercooled liquid sample at a specific undercooling level under carefully controlled conditions. The capabilities of this equipment allow an important new step in the examination of the incorporation behavior by providing access to a wider range and greater number of undercooling levels. A computer model has been developed to perform simulations of the DTA signals measured during the solidification of highly undercooled metal matrix composites. The model has been refined to permit the determination of bounds on the solid-liquid interfacial velocities as a function of time and temperature through the analysis of the experimentally measured DTA signals.

The current studies will provide a database for future space experiments where the incorporation behavior can be examined in the absence of the confounding influence of buoyancy-induced convection. The combination of ground-based studies and experiments under microgravity conditions will provide an understanding of the key parameters of the particle incorporation process that can also lead to improved composite processing strategies for both uniform gradient particle distributions.

Bibliography Type:	Description: (Last Updated: 04/29/2010)
Dissertations and Theses	Sebright JL. "Particle-dendrite interactions during the solidification of undercooled metal-matrix composites." Ph. D. thesis, University of Wisconsin - Madison, May 2004. , May-2004